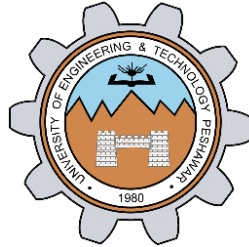


Interfacing Push buttons to 8051 Development kit using Polling

LAB # 03



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Date:

5th March 2024

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Task 1:

Code:

```
STARTUP.A51  Example1.c
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit MyBit = P1^0;
5
6  void main(void){
7
8      unsigned int i;
9
10     for(i = 0; i<=500; i++){
11         MyBit = 0;
12         MyBit = 1;
13     }
14 }
```

Output:

```
11:      MyBit = 0;
C:0x0803  C290  CLR      MyBit(0x90.0
12:      MyBit = 1;
C:0x0805  D290  SETB     MyBit(0x90.0
13:      }
```

Parallel Port 1

Port 1	7	Bits	0
P1: 0xFE	☒	☒	☒
Pins: 0xFE	☒	☒	☒

STARTUP.A51 Example1.c EXAMPL1.C

```
9
10  for(i = 0; i<=500; i++){
11      MyBit = 0;
12      MyBit = 1;
13  }
```

```
11:      MyBit = 0;
C:0x0803  C290  CLR      MyBit(0x90.0
12:      MyBit = 1;
C:0x0805  D290  SETB     MyBit(0x90.0
13:      }
C:0x0807  0F    INC      R7
```

Parallel Port 1

Port 1	7	Bits	0
P1: 0xFF	☒	☒	☒
Pins: 0xFF	☒	☒	☒

STARTUP.A51 Example1.c EXAMPL1.C

```
9
10  for(i = 0; i<=500; i++){
11      MyBit = 0;
12      MyBit = 1;
13  }
```

Task 2:

Code:

```
Example2.c
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit MyBit = P1^0;
5
6  void main(void){
7
8      unsigned int i;
9
10     for(i = 0; i<=500; i++){
11         MyBit = 0;
12         for(i = 0; i<=10000; i++);
13         for(i = 0; i<=10000; i++);
14         for(i = 0; i<=10000; i++);
15         MyBit = 1;
16     }
17 }
```

Output:

```
Example2.c
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit MyBit = P1^0;
5
6  void main(void){
7
8      unsigned int i;
9
10     for(i = 0; i<=500; i++){
11         MyBit = 0;
12         for(i = 0; i<=10000; i++);
13         for(i = 0; i<=10000; i++);
14         for(i = 0; i<=10000; i++);
15         MyBit = 1;
16     }
17 }
```

Parallel Port 1

Port 1	7	Bits	0
P1: 0xFF	☒	☒	☒
Pins: 0xFE	☒	☒	☐

Task 3:

Code:

```
Example3.c  STARTUP.A51
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit led1 = P1^0;
5  sbit switch1 = P1^1;
6
7  void main(void) {
8
9      while(1) {
10
11         if(switch1 == 0) {
12             led1 = 1;
13         }
14
15         else {
16             led1 = 0;
17         }
18     }
19 }
```

Output:

```
Example3.c  STARTUP.A51
5  sbit switch1 = P1^1;
6
7  void main(void) {
8
9      while(1) {
10
11         if(switch1 == 0) {
12             led1 = 1;
13         }
14
15         else {
16             led1 = 0;
17         }
18     }
19 }
```

Parallel Port 1

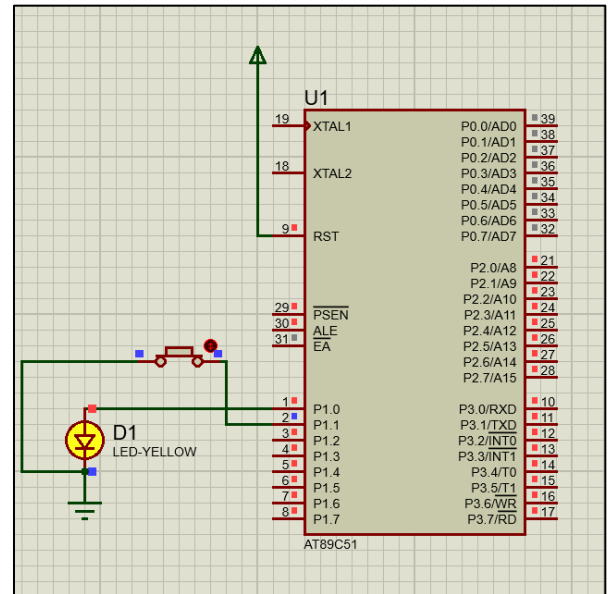
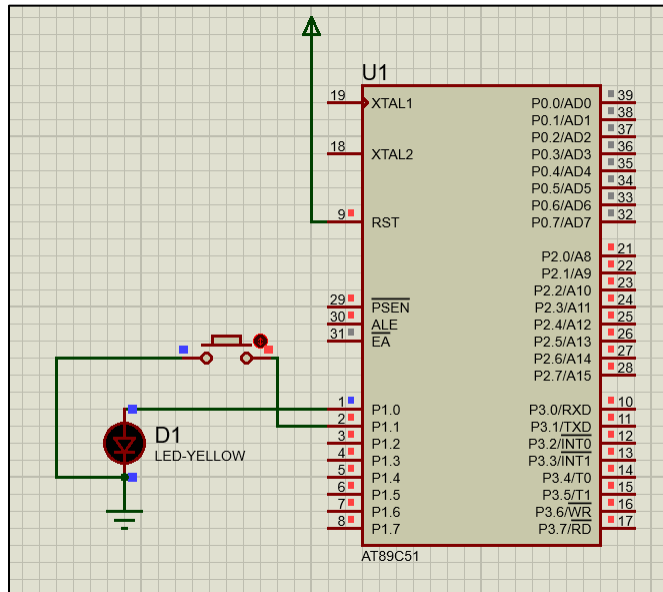
Port 1

P1: 0xFE 7 Bits 0

Pins: 0xFE

7	6	5	4	3	2	1	0
✓	✓	✓	✓	✓	✓	✓	✗

Proteus Simulation:



Task 4:

Code:

```

Example4.c
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit led1 = P1^0;
5  sbit led2 = P1^1;
6  sbit led3 = P1^2;
7  sbit led4 = P1^3;
8
9  sbit switch1 = P2^1;
10 sbit switch2 = P2^2;
11
12 void main(void) {
13
14     while(1) {
15
16         if(switch1 == 1 & switch2 == 1) {
17             led1 = 1;
18             led2 = 0;
19             led3 = 0;
20             led4 = 0;
21         }
    }

```

```

20     led4 = 0;
21 }
22
23 else if(switch1 == 1 & switch2 == 0){
24     led1 = 0;
25     led2 = 1;
26     led3 = 0;
27     led4 = 0;
28 }
29
30 else if(switch1 == 0 & switch2 == 1){
31     led1 = 0;
32     led2 = 0;
33     led3 = 1;
34     led4 = 0;
35 }
36
37 else {
38     led1 = 0;
39     led2 = 0;
40     led3 = 0;
41     led4 = 1;
42 }
43 }
44 }

```

Output:

Example4.c

STARTUP.A51

```

12 void main(void){
13
14     while(1){
15
16         if(switch1 == 1 & switch2 == 1){
17             led1 = 1;
18             led2 = 0;
19             led3 = 0;
20             led4 = 0;
21         }
22
23         else if(switch1 == 1 & switch2 == 0){
24             led1 = 0;
25             led2 = 1;
26             led3 = 0;
27             led4 = 0;

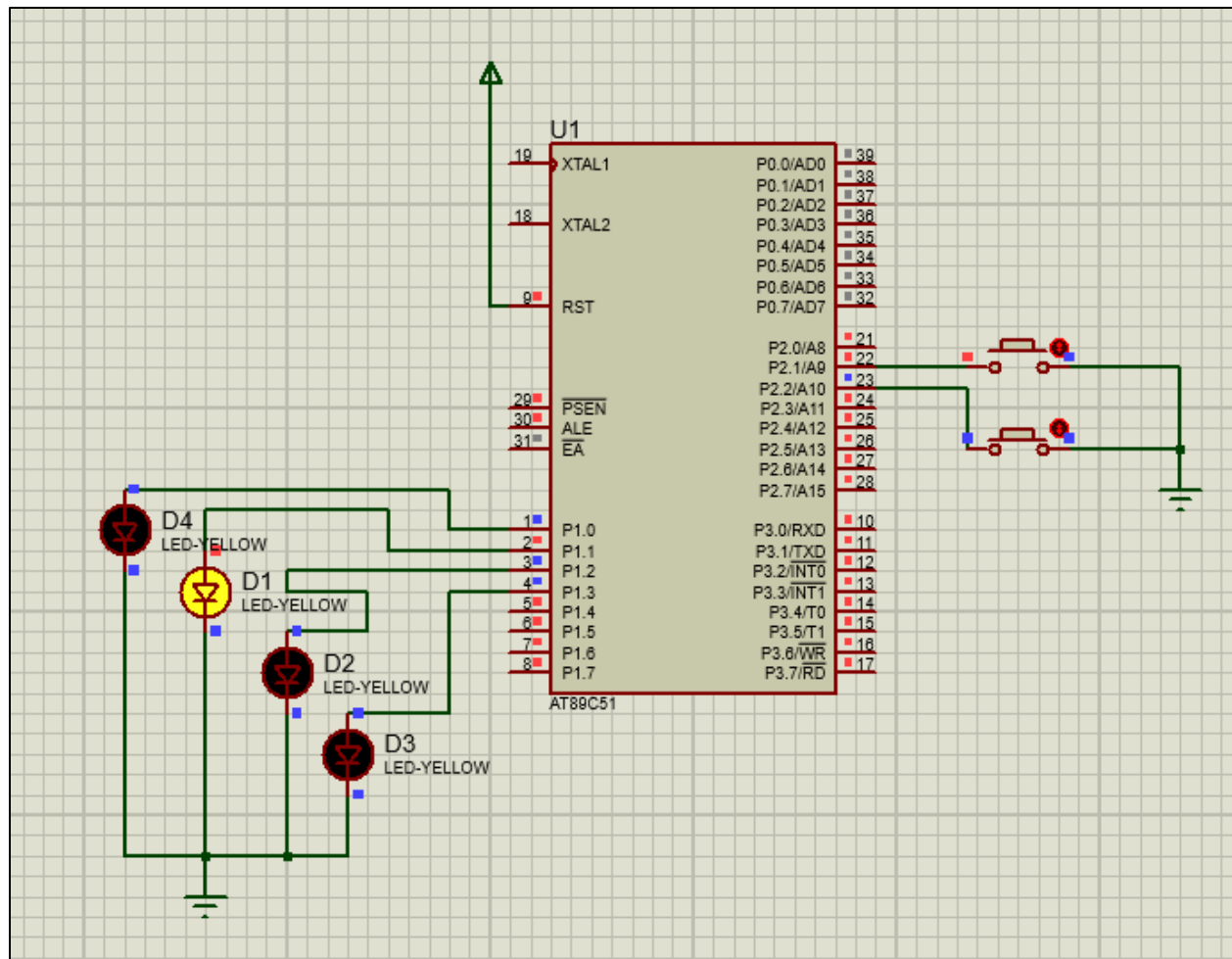
```

Parallel Port 1

Port 1

P1:	0xF1	7	Bits	0
		✓	✓	✓
Pins:	0xF1	✓	✓	✓

Proteus Simulation:



Task 5:

Attach 3 push button with PORT2 and 8 LED's with PORT1 and turn ON led when the corresponding button are pressed, use polling for monitoring the buttons.

Code:

```
Task.c  STARTUP.A51
1  #include<reg51.h>
2  #include<stdio.h>
3
4  sbit switch1 = P2^1;
5  sbit switch2 = P2^2;
6  sbit switch3 = P2^3;
7
8  void main(void) {
9
10     while(1) {
11
12         if(switch1 == 1 & switch2 == 1 & switch3 == 1) {
13             P1 = 0x01;
14         }
15         else if(switch1 == 1 & switch2 == 1 & switch3 == 0) {
16             P1 = 0x02;
17         }
18
19         else if(switch1 == 1 & switch2 == 0 & switch3 == 1) {
20             P1 = 0x04;
21         }
22
23         else if(switch1 == 1 & switch2 == 0 & switch3 == 0) {
24             P1 = 0x08;
25         }
26
27         else if(switch1 == 0 & switch2 == 1 & switch3 == 1) {
28             P1 = 0x10;
29         }
30
31         else if(switch1 == 0 & switch2 == 1 & switch3 == 0) {
32             P1 = 0x20;
33         }
34
35         else if(switch1 == 0 & switch2 == 0 & switch3 == 1) {
36             P1 = 0x40;
37         }
38
39         else {
40             P1 = 0x80;
41         }
42     }
43 }
```


Output:

```
Task.c  STARTUP.A51
9
10 while(1){
11
12     if(switch1 == 1 & switch2 == 1 & switch3 ==
13         P1 = 0x01;
14 }
15 else if(switch1 == 1 & switch2 == 1 & switch3 == 0){
16     P1 = 0x02;
17 }
18
19 else if(switch1 == 1 & switch2 == 0 & switch3 == 1){
```

Parallel Port 1

Port 1

P1: 0x01 7 Bits 0

Pins: 0x01

Proteus Simulation:

